EESC4550/5550: Practical Atmospheric Modeling (Fall 2026)
Lab 3 – The Community Earth System Model
Total Score: 20

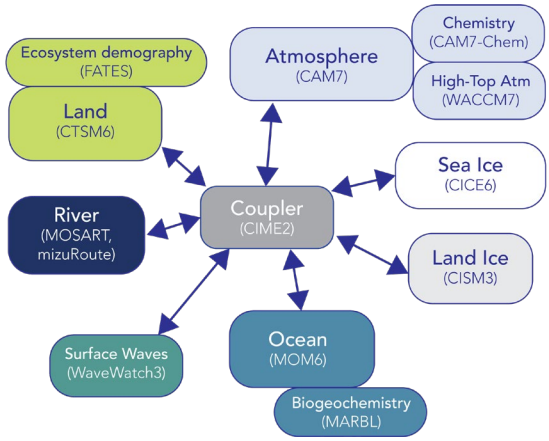
Learning Objectives:

- Build a new case based on cloning from an existing one.
- Practice the running a simulation on CESM and identify where start date, end date, run length, output frequency, and output duration are controlled.
- Run a short, coupled simulation and interpret its output files.
- Answer conceptual questions about the structure of CESM.

In this lab, we will setup the Community Earth System Model (CESM 2.2.2).

Architecture of CESM

CESM is a fully coupled, state-of-the-art climate model that simulates the Earth's atmosphere, ocean, land surface, sea ice, land ice, and biogeochemical cycles as a single interacting system.



Component	Acronym	Full name
Atmosphere	CAM	Community Atmosphere Model
Ocean	POP	Parallel Ocean Program
Sea Ice	CICE	Community Ice CodE
Land	CLM	Community *Land* Model
River	MOSART	Model for Scale Adaptive *River* Transport
Land Ice	CISM	Community Ice *Sheet* Model
Wave	WW3	WAVEWATCH III
Coupler	CPL	Coupler (regrids and exchanges fluxes between components)

One-Time Setup

1. Connecting to the Compute Node and Clearing the Environment

```
export SID=$(whoami)
ssh sandbox
srun --partition=chpc --nodes=1 --ntasks=1 --cpus-per-task=1 \
     --time=00:30:00 --account=eesc4550 --pty bash
conda deactivate   # peacefully ignore the error if conda is unrecognized
module purge
```

2. Copying CESM Configuration Files Defined for the CUHK Central Cluster and Activating the Python Environment

```
mkdir -p $HOME/.cime
cp /lustre/project/eesc4550/shared/cime_config/* $HOME/.cime/
chmod u+w $HOME/.cime/*
source /lustre/project/eesc4550/cesm_env/bin/activate
```

3. Cloning the Master Case and Changing the Path Settings

```
export SID=$(whoami)
cd /lustre/project/eesc4550/CESM/cime/scripts
./create_clone --clone /lustre/project/eesc4550/cases/ssp585_2026 \
               --case /lustre/project/eesc4550/$SID/cases/ssp585 \
       --cime-output-root /lustre/project/eesc4550/$SID/cases/ssp585/cesm_scratch
cd /lustre/project/eesc4550/$SID/cases/ssp585
./xmlchange CIME_OUTPUT_ROOT=/lustre/project/eesc4550/$SID/cases/ssp585/cesm_scratch
./xmlchange EXEROOT=/lustre/project/eesc4550/$SID/cases/cesm_scratch/ssp585/bld
./xmlchange RUNDIR=/lustre/project/eesc4550/$SID/cases/cesm_scratch/ssp585/run
./xmlchange DOUT_S_ROOT=/lustre/project/eesc4550/$SID/cases/cesm_archive/ssp585
```

4. Building the Model and Submitting a Task, Then Exit from the Compute Node

```
./case.setup --clean
./case.setup
./case.build --clean
./case.build --skip-provenance-check
exit
```

Per-Simulation Setup

```
export SID=$(whoami)
srun --partition=chpc --nodes=1 --ntasks=1 --cpus-per-task=1 \
     --time=00:30:00 --account=eesc4550 --pty bash
conda deactivate
module purge
source /lustre/project/eesc4550/cesm_env/bin/activate
cd /lustre/project/eesc4550/$SID/cases/ssp585
./case.submit
squeue
exit
```

Checking the Output

```
cd /lustre/project/eesc4550/$SID/cases/ssp585/cesm_scratch
```

Answer the following questions.

Question 1 (5 pts). Provide **THREE** screenshots, one to show that you have built the CESM model successfully, one to show that the model is running, and one to show that you have successfully obtained the simulation results.

Question 2 (3 pts). List at least two core differences between the Community Earth System Model (CESM) and GEOS-Chem. What are the considerations for adopting GEOS-Chem for a research study rather than using CESM?

Question 3 (2 pts). Your `run` directory contains two types of `.nc` files from the atmosphere component, `ssp585.cam.r.2026-01-03-00000.nc` and `ssp585.cam.rh0.2026-01-03-00000.nc`.

(a) What is the difference between an `r` file and an `rh0` file?

(b) The `rpointer.atm` file in your run directory contains a path. What is that path, and what does it tell the model to do on the next run?

Question 4 (3 pts). You have already run a 2-day simulation starting on 2026-01-01. You want to simulate 48 hours of atmosphere-ocean interaction starting at midnight on 2026-06-15.

(a) Which settings in the case directory do you need to change? Describe how the changes should be made.

(b) After you submit and the run finishes, what is the calendar date and time of the final restart file? Write it in the filename format used by CESM: `ssp585.<component>.r.YYYY-MM-DD-<seconds>.nc`

Question 5 (3 pts). Inspect the atmosphere history file and look at the time dimension. How many samples are stored in the file, and how frequently did CAM write to the history file? Add a line to `user_nl_cam` so that the primary history stream is written **every 6 hours**. What values do you set for `nhtfrq` and `mfilt`?

Question 6 (4 pts). In this lab, the master case uses the compset: `SSP585_CAM60_CLM50%BGC-CROP_CICE_POP2%ECO%ABIO-DIC_MOSART_CISM2%NOEVOLVE_WW3_BGC%BDRD` and the res: `f19_g17`.

(a) The `SSP585` prefix indicates the emission scenario. What does SSP5-8.5 represent, and why is it used as a "high-end" scenario?

(b) The `%BGC-CROP` and `%ECO%ABIO-DIC` suffixes are modifiers. What do they change about the default CLM and POP configurations?

(c) What exactly is `f19_g17`?

(d) If you wanted to run a **pre-industrial, atmosphere-only** simulation (no ocean, no sea ice), what would the compset look like? Name the specific compset you would use.

Submission Guideline: To complete the lab, please submit **ONE** PDF file:

- `lab3_<your_id>.pdf`, which contains the answers to the **6** questions.

----- **End of Lab 3** -----